

6(a)(i)	<i>any three from</i> haem group containing iron (atom) ; four haem groups (per molecule of haemoglobin) ; each iron (atom) / haem / prosthetic group, binds one oxygen molecule ; or four oxygen molecule per haemoglobin molecule ; cooperative binding / allosteric effect described ; AVP ;	3
6(a)(ii)	<i>any two from</i> <i>ref.</i> folding of polypeptide chain(s) / globin(s) fold (to give globular structure) ; A <i>ref.</i> to, tertiary structure / quaternary structure (so) hydrophilic R groups (of amino acids) are on the outside of the molecule / hydrophobic R groups (of amino acids) on the inside of the molecule ; A hydrophilic and hydrophobic amino acids hydrogen bonds form with water (molecules) ;	2
6(b)(i)	llama = 90.5% human = 78.5% ;	1
6(b)(ii)	<i>any two from</i> lower partial pressure of oxygen at high altitude ; at the same partial pressure the percentage saturation is (much) higher than for humans ; <i>ref.</i> to haemoglobin has a higher affinity for oxygen ; <i>idea that</i> sufficient oxygen delivered to tissues to satisfy demand ;	2